Java Collections – Maps and Set Applications

Assignment 01: Student Marks Lookup (HashMap)

Objective

Store student names and marks for quick lookup operations.

Tasks

1. Accept student names and marks from the user.
2. Store the records using `HashMap`.
3. Search marks using a student name.
4. Display the marks of the searched student.

Concepts Used

- HashMap
- Key-Value Mapping
- Fast Lookup Operations

Assignment 02: Bank Account Directory (HashMap)

Objective

Maintain a bank account directory using account numbers and customer names.

Tasks

1. Store account number and customer name pairs.
2. Search customer details using account number.
3. Delete an account from the directory.
4. Display the updated directory.

Concepts Used

- HashMap
- CRUD Operations on Maps
- EntrySet Traversal

Assignment 03: Employee Sorting System (TreeMap)

Objective

Store employee records sorted by employee ID.

Tasks

1. Accept employee ID and employee name entries.
2. Store records in a `TreeMap`.
3. Display employees in ascending order of IDs.

Concepts Used

- TreeMap
- Automatic Sorting
- Map Traversal

Assignment 04: City Temperature Report (TreeMap)

Objective

Generate a sorted city temperature report and identify hottest/coldest cities.

Tasks

1. Store city names with temperatures.

2. Display cities in sorted order.
3. Find the hottest city.
4. Find the coldest city.

Concepts Used

- TreeMap
- Sorted Data Storage
- Maximum and Minimum Search

Assignment 05: Order History Tracker (LinkedHashMap)

Objective

Maintain customer order history while preserving insertion order.

Tasks

1. Store order IDs and product names.
2. Display order history in insertion order.
3. Remove a cancelled order.
4. Display updated order history.

Concepts Used

- LinkedHashMap
- Insertion Order Preservation
- Map Manipulation

Assignment 06: System Log Tracker (LinkedHashMap)

Objective

Store system logs while maintaining insertion order.

Tasks

1. Accept multiple log entries.
2. Store logs using `LinkedHashMap`.
3. Display logs in insertion order.

Concepts Used

- LinkedHashMap
- Ordered Data Storage

Assignment 07: Online Voting System (HashSet + HashMap)

Objective

Prevent duplicate voting and maintain candidate vote counts.

Tasks

1. Store voter IDs to prevent duplicate voting.
2. Maintain candidate vote counts.
3. Reject duplicate votes.
4. Display final voting results.

Concepts Used

- HashSet
- HashMap
- Duplicate Prevention
- Vote Counting System

Assignment 08: Unique Number Counter (HashSet)

Objective

Count unique numbers from user input.

Tasks

1. Accept N integers from the user.
2. Store them in a `HashSet`.
3. Display the count of unique numbers.

Concepts Used

- HashSet
- Duplicate Elimination

Assignment 09: College Admission System (HashSet)

Objective

Store unique student application IDs.

Tasks

1. Accept multiple application IDs.
2. Store IDs using `HashSet`.
3. Display unique application IDs.

Concepts Used

- HashSet
- Unique Data Storage

Assignment 10: Word Frequency Counter (HashMap + HashSet)

Objective

Count frequency of words in a sentence.

Tasks

1. Split a sentence into words.
2. Identify unique words.
3. Count frequency of each word.
4. Display frequency report.

Concepts Used

- HashMap
- HashSet
- Frequency Counting
- String Processing